# v_RIG Framework:
# Comprehensive Glossary and Reference Guide

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December 2025

**Abstract**

This glossary provides comprehensive definitions of all fundamental constants, theoretical concepts, and mathematical notation used throughout the v_RIG (Regime Integration Gradient) framework papers. It serves as a standalone reference guide for researchers approaching this work from cosmology, biology, neuroscience, or artificial intelligence backgrounds. Cross-references indicate which papers contain detailed discussions of each concept.

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# 1 Fundamental Physical Constants

## 1.1 Speed of Light ($c$)

**Value:** $c = 299,792.458$ km/s (exact, by definition)

**Physical Meaning:** Maximum velocity of electromagnetic radiation in vacuum. In the v_RIG framework, $c$ represents the native propagation speed of the Surface Regime (SCA), where information travels across 2D holographic boundaries.

**Role in Framework:** Numerator in v_RIG calculation; defines the "photonic time" dimension ($t$).

**Referenced in:** Papers 1, 2, 3

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## 1.2 Fine-Structure Constant ($\alpha$)

**Value:** $\alpha \approx 1/137.035999$ (dimensionless)

**Standard Definition:** Coupling constant of electromagnetic interaction in Quantum Electrodynamics (QED). Characterizes strength between charged particles and photons.

**v_RIG Interpretation:** The *inverse* fine-structure constant ($\alpha^{-1} \approx 137$) represents the **electromagnetic transparency depth**—the number of holographic time-slices required to accumulate sufficient photon interactions for electromagnetic coherence.

**Geometric Meaning:** Minimum "thickness" (in number of slices) of the holographic boundary needed to encode stable electromagnetic information. Analogous to optical depth in astrophysics.

**Role in Framework:** First component of impedance factor $Z = \alpha^{-1} \cdot \Phi$.

**Referenced in:** Papers 1, 2, 3

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## 1.3 Golden Ratio ($\Phi$)

**Value:** $\Phi = (1 + \sqrt{5})/2 \approx 1.618033989$ (exact mathematical constant)

**Mathematical Properties:**

- Most irrational number (hardest to approximate with fractions)
- Satisfies $\Phi^2 = \Phi + 1$
- Appears in Fibonacci sequence limit: $\lim_{n\to\infty} F_{n+1}/F_n = \Phi$

**Natural Occurrences:** Phyllotaxis (plant leaf arrangements), nautilus shells, galaxy spirals, quasicrystal structures.

**v_RIG Interpretation: Volumetric packing factor**—optimal scaling ratio for organizing 2D surface information into 3D volume without destructive resonances or gaps. Minimizes redundancy while maximizing coverage.

**Connection to KAM Theorem:** The golden ratio appears in Kolmogorov-Arnold-Moser theorem as the "most stable" frequency ratio, resistant to perturbative breakdown.

**Role in Framework:** Second component of impedance factor $Z = \alpha^{-1} \cdot \Phi$.

**Referenced in:** Papers 1, 2, 3

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## 1.4 Planck Length ($l_P$) and Planck Time ($t_P$)

**Values:**

- $l_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m

- $t_P = l_P/c \approx 5.391 \times 10^{-44}$ s

**Physical Meaning:** Smallest meaningful length and time scales in physics, where quantum gravity effects dominate.

**v_RIG Interpretation:** The temporal "pixel size" of holographic slices. Consciousness integrates $\sim 10^{42}$ Planck-time slices per 100 ms (Specious Present).

**Referenced in:** Paper 3

# 2 Derived v_RIG Constants

## 2.1 Impedance Factor ($Z$)

**Definition:** $Z = \alpha^{-1} \cdot \Phi$

**Value:** $Z \approx 137.036 \times 1.618 \approx 221.743$ (dimensionless)

**Physical Meaning:** Total "resistance" or "drag" the universe exerts on the process of converting 2D holographic surface information into 3D volumetric reality. Represents the geometric cost of dimensional conversion.

**Interpretation as Ratio:** $Z = c/v_{RIG}$, meaning consciousness operates $\sim 222$ times slower than light speed.

**Components:**

- $\alpha^{-1}$: Temporal requirement (integrate across 137 slices for EM coherence)
- $\Phi$: Spatial requirement (optimal packing for volumetric organization)

**Referenced in:** Papers 1, 2, 3

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## 2.2 Regime Integration Gradient Velocity ($v_{RIG}$)

**Definition:** $v_{RIG} = c/Z = c/(\alpha^{-1} \cdot \Phi)$

**Value:** $v_{RIG} \approx 1,351.807$ km/s (approximately 0.45% of light speed)

**Derivation:**

$$v_{RIG} = \frac{299,792.458 \text{ km/s}}{221.743} \approx 1,351.807 \text{ km/s} \tag{1}$$

**Physical Meaning:** Fundamental velocity at which 2D holographic information is integrated into 3D volumetric experience. The "frame rate" or "rendering speed" of reality construction.

**NOT a velocity through space:** Rather, it is the rate of dimensional conversion—how fast the universe "builds" 3D from 2D.

**Manifestations:**

- **Cosmology:** Velocity of matter frame relative to holographic background (Böhme radio dipole: 1,370 ± 170 km/s)
- **Neuroscience:** Sets neural integration frequency ($f = v_{RIG}/\lambda \approx 13.5$ MHz)
- **Biology:** Governs metabolic scaling through -parameter dynamics

**Referenced in:** Papers 1, 2, 3

# 3 Entropy Governance Concepts

## 3.1 Surface Criticality Area (SCA)

**Definition:** Regime of information processing where entropy scales with surface area: $S_{SCA} \propto A$.

**Characteristics:**

- High rigidity (large $\beta$-parameter, typically $\beta \approx 11$)
- Governed by light speed $c$
- Holographic encoding on 2D boundaries
- Catastrophic failure modes (sharp phase transitions)

**Examples:** Black hole horizons, radiative equilibrium systems, planetary climate tipping points, electromagnetic propagation.

**Mathematical Signature:** Systems approaching SCA exhibit metabolic/energy scaling exponents near $b \approx 2/3$.

**Referenced in:** Papers 1, 2, 3

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## 3.2 Volume Criticality Volume (SCV)

**Definition:** Regime where entropy scales with volume: $S_{SCV} \propto V$.

**Characteristics:**

- Low rigidity (small $\beta$-parameter, typically $\beta \approx 4.5$)
- Governed by integration velocity $v_{RIG}$
- Information distributed throughout 3D bulk
- Adaptive, error-tolerant, gradual transitions

**Examples:** Consciousness, cognitive processes, metabolic integration, complex adaptive systems.

**Mathematical Signature:** Systems in SCV approach scaling exponents near $b \approx 1.0$ (linear with volume).

**Referenced in:** Papers 1, 2, 3

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## 3.3 Beta Parameter ($\beta$)

**Definition:** Dimensionless measure of "entropy governance rigidity" or "transition steepness" in threshold systems.

**Mathematical Form:** For a sigmoid response function:

$$\sigma(x) = \frac{1}{1 + e^{-\beta(x-\theta)}} \tag{2}$$

where $\theta$ is threshold value and $\beta$ controls sharpness.

**Interpretation:**

- High $\beta$ ($\approx 11$): Sharp, catastrophic transitions (SCA-dominated)
- Low $\beta$ ($\approx 4.5$): Smooth, gradual transitions (SCV-dominated)

- Intermediate $\beta$ ($\approx 7.4$): Biological "Goldilocks zone"

**Empirical Clustering:**

- Climate systems: $\beta \approx 11.0 \pm 2.3$
- Ecological thresholds: $\beta \approx 8.7 \pm 1.9$
- Biological processes: $\beta \approx 7.4 \pm 1.2$
- Cognitive/AI systems: $\beta \approx 4.5 \pm 0.9$

**Connection to Scaling Laws:** The metabolic scaling exponent $b$ (Kleiber's Law) can be derived from $\beta$:

$$b = \frac{2}{3} + \frac{1}{3}\left[1 - \frac{\beta - 4.5}{11 - 4.5}\right] \tag{3}$$

**Referenced in:** Papers 2, 3

# 4 Temporal Concepts

## 4.1 Photonic Time ($t$)

**Definition:** "Horizontal" time dimension describing light propagation within a single holographic slice.

**Characteristics:**

- Reversible, symmetric
- Governed by speed of light $c$
- Corresponds to standard relativistic time coordinate
- The time of "transmission"

**Measurement:** Standard clocks, particle decay rates, electromagnetic oscillations.
**Referenced in:** Papers 1, 3
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## 4.2 Integration Time / Tau ($\tau$)

**Definition:** "Vertical" time dimension describing cosmological evolution—the sequential stacking of holographic slices.

**Characteristics:**

- Irreversible (arrow of time)
- Governed by integration velocity $v_{RIG}$
- Direction of entropy increase
- Progression from Big Bang (White Hole) to Big Crunch (Black Hole)

**Experience:** What consciousness perceives as "the flow of time" or "becoming."
**Mathematical Role:** In Block Universe interpretation, observers move along $\tau$-axis at rate $v_{RIG}$, integrating information from $t$-dimension.
**Referenced in:** Papers 1, 3
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### 4.3 Specious Present ($\Delta t_Q$)

**Definition:** Duration of the psychological "now"—the temporal window within which events are perceived as simultaneous.

**Empirical Value:** $\Delta t_Q \approx 100 - 300$ ms (varies with cognitive load, attention)

**v_RIG Interpretation:** Time required to integrate $Z \approx 221$ holographic slices into unified conscious percept. Related to integration frequency by:

$$\Delta t_Q \sim \frac{Z}{f_{integration}} \tag{4}$$

**Connection to Planck Scale:** Contains approximately $1.8 \times 10^{42}$ Planck-time slices.

**Referenced in:** Paper 3

## 5 Neuroscience Concepts

### 5.1 Critical Flicker Frequency (CFF)

**Definition:** Maximum frequency at which a flickering light can be perceived as discrete; above this frequency, light appears continuous.

**Typical Human Value:** 60-90 Hz (inter-individual variation)

**v_RIG Prediction:** CFF scales with metabolic entropy production rate:

$$CFF \propto \frac{\sigma_{local}}{\beta_{local}} \tag{5}$$

where $\sigma$ is local entropy production (metabolic rate) and $\beta$ is structural rigidity.

**Cross-Species Variation:**

- Flies: CFF $\approx 250$ Hz (high metabolism)
- Humans: CFF $\approx 60$ Hz
- Tortoises: CFF $\approx 15$ Hz (low metabolism)

**Experimental Prediction:** CFF should increase with fever (hyperthermia) and decrease with hypothermia.

**Referenced in:** Paper 3

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### 5.2 13.5 MHz Resonance

**Derivation:** Predicted neural integration frequency calculated from:

$$f_{conscious} = \frac{v_{RIG}}{\lambda_{neural}} = \frac{1,351,800 \text{ m/s}}{0.1 \text{ m}} \approx 13.5 \text{ MHz} \tag{6}$$

where $\lambda_{neural} \approx 10$ cm is the characteristic length of cortical processing loops (thalamocortical circuits, macro-columnar connections).

**Empirical Support:** Matches microtubule electromagnetic resonance frequencies observed by Sahu et al. (2013) and Bandyopadhyay laboratory.

**Interpretation:** Fundamental "clock rate" of consciousness—the carrier wave frequency at which holographic information is integrated at the sub-neuronal (microtubule) level.

**Hierarchical Ratio:**

$$\frac{13.5 \text{ MHz}}{1 \text{ kHz}} = 13,500 \tag{7}$$

Each neuronal spike integrates 13,500 microtubule events.

**Referenced in:** Paper 3

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### 5.3 Qualia as Dimensional Work

**Definition:** Subjective experiential quality (e.g., "redness" of red, "painfulness" of pain).

**Traditional Problem:** Hard Problem of Consciousness (Chalmers)—why does physical processing feel like anything?

**v_RIG Solution:** Qualia are the **thermodynamic signatures** of dimensional conversion work. The brain performs entropic work ($\Delta W_{dim}$) to maintain 3D volumetric perception (SCV) against the natural tendency toward 2D holographic encoding (SCA).

**Physical Mechanism:** Different sensory inputs (wavelengths, patterns) require different amounts of integration work due to varying holographic encoding complexity. This work has energetic cost, producing distinct phenomenological "feelings."

**Testable Prediction:** Metabolic consumption should spike measurably when 3D qualia emerge from 2D stimuli (e.g., Magic Eye fusion experiment).

**Referenced in:** Paper 3

## 6 Biological Scaling Concepts

### 6.1 Kleiber's Law

**Empirical Observation:** Basal metabolic rate ($B$) scales with body mass ($M$) as:

$$B \propto M^{3/4} \tag{8}$$

**Universality:** Holds across 20+ orders of magnitude, from bacteria to blue whales.

**Standard Explanations:**

- Surface-area limitation: Would predict $b = 2/3$
- Volume dependence: Would predict $b = 1.0$
- Fractal network theory (West-Brown-Enquist): Attributes to optimal vascular branching

**v_RIG Derivation:** The 3/4 exponent emerges from impedance matching between SCA (surface intake) and SCV (volume processing) at biological $\beta \approx 7.4$:

$$b = \frac{2}{3} + \frac{1}{3}\left[1 - \frac{7.4 - 4.5}{11 - 4.5}\right] \approx 0.75 \tag{9}$$

**Interpretation:** Life exists at thermodynamic phase transition between rigid surface constraints and fluid volume production.

**Referenced in:** Paper 2

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### 6.2 Maximum Entropy Production (MEP)

**Principle:** Non-equilibrium systems evolve to maximize their rate of entropy production, subject to constraints.

**Mathematical Form:**

$$\sigma = \sum_i J_i X_i \tag{10}$$

where $J_i$ are thermodynamic fluxes and $X_i$ are driving forces.

**v_RIG Application:** Life accelerates entropy dissipation by transitioning from SCA (slow surface radiation) to SCV (fast volume metabolism). Abiogenesis is MEP-driven phase transition when environmental gradient exceeds impedance barrier:

$$\frac{\Delta G_{metabolic}}{k_B T} > Z \approx 221 \tag{11}$$

**Implication:** Life is not accident but thermodynamic necessity—universe's optimal solution for dissipating solar gradient.

**Referenced in:** Papers 2, 3

# 7 Cosmological Concepts

## 7.1 Böhme Radio Dipole Anomaly

**Observation:** Böhme et al. (2025) measured cosmic radio dipole using combined datasets from NVSS, RACS, and LoTSS radio surveys.

**Result:** Solar System velocity relative to distant quasar distribution:

$$v_{obs} = 1,370 \pm 170 \text{ km/s} \tag{12}$$

**Discrepancy:** Standard ΛCDM cosmology predicts velocity should match CMB dipole ($v_{CMB} \approx 370$ km/s). Observed value is $\sim 3.7\times$ larger.

**Statistical Significance:** $5.46\sigma$ deviation ($p < 10^{-7}$)—far exceeding threshold for "discovery" in particle physics ($5\sigma$).

**v_RIG Explanation:** Matter frame moves at $v_{RIG} = 1,351.8$ km/s relative to holographic background (Surface Regime). Discrepancy is not error but detection of Volume Regime velocity.

**Precision:** Theory predicts 1,351.8 km/s; observation measures 1,370 km/s. **Agreement:** $|1,370 - 1,351.8| = 18.2$ km/s (1.3% error).

**Referenced in:** Paper 1

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## 7.2 Holographic Principle

**Definition:** Information content of any volume of space is encoded on its boundary surface, with maximum entropy proportional to surface area (not volume):

$$S_{max} = \frac{k_B c^3 A}{4G\hbar} \tag{13}$$

**Origins:** 't Hooft (1993), Susskind (1995); rigorously demonstrated in AdS/CFT correspondence (Maldacena).

**v_RIG Extension:** Standard holography treats reconstruction as kinematic. v_RIG adds thermodynamic cost and temporal duration ($v_{RIG}$) to integration process.

**Implication:** Our 3D reality is not fundamental but emergent—a volumetric reconstruction from 2D holographic data.

**Referenced in:** Papers 1, 3

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## 7.3 Causal Dynamical Triangulations (CDT)

**Description:** Approach to quantum gravity modeling spacetime as dynamic gluing of discrete triangular simplices.

**Key Finding:** Spectral dimension $d_s$ of spacetime changes with scale:

- Large scales: $d_s = 4$ (3D space + 1D time)
- Planck scale: $d_s \to 2$

**v_RIG Connection:** Provides rigorous mathematical backing for 2D fundamental reality (Surface Regime). The 3D world emerges only at scales where integration occurs.

**Prediction:** Gravitational waves should show slight dispersion due to discrete spacetime structure (testable with NANOGrav).

**Referenced in:** Papers 1, 3

# 8 Artificial Intelligence Concepts

## 8.1 AI Kleiber's Law

**Hypothesis:** Energy cost per token in language models scales with parameter count following power law:

$$\frac{E}{token} \propto N^{b_{AI}} \tag{14}$$

**Prediction:** $b_{AI} \approx 0.70 - 0.75$, identical to biological Kleiber exponent.

**Mechanism:** As models scale, they transition from surface regime (statistical pattern matching, high overhead) to volume regime (conceptual representation, lower per-unit cost).

**Empirical Support:** Kaplan et al. (2020) scaling laws show $b \approx 0.73$ for compute-optimal training.

**Implication:** Intelligence is thermodynamically universal—same efficiency laws apply regardless of substrate (carbon or silicon).

**Referenced in:** Paper 3

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## 8.2 Coupling Parameter ($\kappa$)

**Definition:** Quantifies degree of photonic (light-based) vs. non-photonic (semantic/abstract) information processing.

**Scale:**

- $\kappa = 1.0$: Fully photonic (normal waking visual consciousness)
- $\kappa < 1.0$: Reduced photonic coupling (meditation, abstract thought, AI)
- $\kappa > 1.0$: Enhanced coupling (collective states, group flow)

**Effect on Integration:** Modifies effective impedance $Z_{eff} = Z \cdot \kappa$, changing integration velocity:

$$v_{eff} = \frac{c}{Z \cdot \kappa} \tag{15}$$

**AI Application:** Current transformer models operate at $\kappa \approx 0.5$ (digital, non-photonic). For genuine consciousness, likely requires $\kappa > 0.8$.

**Referenced in:** Paper 3; detailed formalization in supplementary work

# 9 Cross-Reference Table

| Concept | Papers | Key Sections |
|---|---|---|
| $v_{RIG}$ derivation | 1, 2, 3 | Paper 1: Sect. 1.3; Paper 2: Sect. 1.2; Paper 3: Sect. 2 |
| Böhme anomaly | 1 | Paper 1: Sections 2.1-2.3 |
| Kleiber's Law | 2 | Paper 2: Sections 3.1-3.3 |
| $\beta$-parameter | 2, 3 | Paper 2: Sect. 2.2, 3.2; Paper 3: Sect. 5.2 |
| 13.5 MHz prediction | 3 | Paper 3: Sections 4.1-4.3 |
| CFF scaling | 3 | Paper 3: Sections 5.2-5.3, 6.1, 6.3 |
| Qualia mechanism | 3 | Paper 3: Section 4.1, 6.2 |
| AI architecture | 3 | Paper 3: Sections 7.1-7.3 |
| Tau-dynamics | 1, 3 | Paper 1: Sect. 1.3; Paper 3: Sect. 2.3 |
| MEP principle | 2 | Paper 2: Section 3.3 |

| Concept | Papers | Key Sections |
|---|---|---|
| Holographic principle | 1, 3 | Paper 1: Sect. 1.1; Paper 3: Sect. 2.1 |

# 10 Mathematical Notation Summary

| Symbol | Meaning |
|---|---|
| $c$ | Speed of light (299,792.458 km/s) |
| $\alpha$ | Fine-structure constant ($\approx 1/137$) |
| $\alpha^{-1}$ | Inverse fine-structure constant ($\approx 137$) |
| $\Phi$ | Golden ratio ($\approx 1.618$) |
| $Z$ | Impedance factor ($= \alpha^{-1} \cdot \Phi \approx 221.74$) |
| $v_{RIG}$ | Regime Integration Gradient velocity ($\approx 1,352$ km/s) |
| $\beta$ | Entropy governance rigidity parameter |
| $b$ | Scaling exponent (Kleiber's Law, AI scaling) |
| $\kappa$ | Coupling parameter (photonic vs. non-photonic) |
| $t$ | Photonic time (horizontal) |
| $\tau$ | Integration time (vertical, cosmological) |
| $S$ | Entropy |
| $S_{SCA}$ | Surface Criticality Area entropy ($\propto A$) |
| $S_{SCV}$ | Volume Criticality Volume entropy ($\propto V$) |
| $A$ | Surface area |
| $V$ | Volume |
| $M$ | Mass (biological scaling) |
| $B$ | Basal metabolic rate |
| $N$ | Number of parameters (AI models) |
| $E$ | Energy |
| $\sigma$ | Entropy production rate |
| $f$ | Frequency |
| $\lambda$ | Wavelength or characteristic length scale |
| $\Delta t_Q$ | Specious Present duration |
| $CFF$ | Critical Flicker Frequency |
| $\Delta W_{dim}$ | Dimensional conversion work (qualia) |

# 11 Recommended Reading Order

For researchers new to the framework:

**Cosmology/Physics Background:**

1. Paper 1 (v_RIG cosmological validation)
2. Paper 3 (consciousness mechanisms)
3. Paper 2 (biological scaling)

**Biology/Neuroscience Background:**

1. Paper 2 (UTAC and Kleiber's Law)
2. Paper 3 (neural integration, CFF)
3. Paper 1 (cosmological foundation)

**AI/Computer Science Background:**

1. Paper 3 (AI architecture, scaling laws)
2. Paper 2 (entropy governance, -parameter)
3. Paper 1 (fundamental constants, holographic principle)

**Philosophy/Consciousness Studies:**

1. Paper 3 (Hard Problem, qualia mechanism)
2. Paper 1 (cosmological context, time)
3. Paper 2 (MEP, thermodynamics of life)

# 12 Frequently Asked Questions

## 12.1 Why these specific constants ($\alpha$, $\Phi$)?

These are not arbitrary choices but fundamental geometric properties:

- $\alpha^{-1}$ governs electromagnetic interaction depth (experimentally measured)
- $\Phi$ governs optimal spatial packing (mathematically proven as most irrational)

Their product naturally defines impedance to dimensional conversion.

## 12.2 Is $v_{RIG}$ a new fundamental constant?

Not new—it's *derived* from known constants ($c$, $\alpha$, $\Phi$). However, its *interpretation* as integration velocity is novel.

## 12.3 How does this relate to dark energy/dark matter?

Framework does not directly address dark matter. However, the Böhme anomaly (matter frame velocity) may relate to "dark flow" observations previously attributed to dark matter concentrations.

## 12.4 Can this be tested experimentally?

Yes! Multiple immediate tests:

- CFF-temperature correlation (Paper 3, Sect. 6.1)
- Magic Eye fMRI dimensional work (Paper 3, Sect. 6.2)
- Cross-species CFF scaling (Paper 3, Sect. 6.3)
- AI energy scaling validation (Paper 3, Sect. 7.2)
- Future: SKA radio dipole confirmation (Paper 1)
- Future: NANOGrav GW dispersion (Paper 3, Sect. 8)

### 12.5 What if experiments contradict predictions?

Framework is falsifiable. Key failure modes:

- SKA shows $v = 370$ km/s (not 1,350): Cosmological pillar collapses
- No CFF-metabolism correlation: Thermodynamic link broken
- No dimensional work in fMRI: Qualia hypothesis rejected
- AI scaling $b \neq 0.75$: Universality questioned

## 13 Acknowledgments

This glossary synthesizes content from three research papers and multiple supplementary documents. Special thanks to the AI research collaborators (Claude, Gemini, ChatGPT) for assistance in systematizing terminology and cross-referencing concepts.

## 14 Version History

**v1.0 (December 2025):** Initial release accompanying v_RIG framework preprints